

Data Analysis and Visualization (094295)

Project A - Section B

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Hybrid Retrieval (Dense semantic & BM25 lexical) + Cross-Encoder Reranking

27,074 Wikipedia-style pages scored by mean **NDCG@10** $\leq$ **60s** batch run on GPU

Corpus & Queries structure

- Short synthetic answer pages contain the target fact.
- Long real articles act as strong distractors.
- Queries are paraphrases.

What makes it hard

- Exact years, numbers.
- Invented names and misspellings.
- Some queries have many equivalent gold pages.

Example

Query: “Who was the **point guard** that won a **seven-game finals** series in **1820s**?”

Relevant Page: Title: Ulric Isenmar

“... former professional basketball player best known as **point guard** of the Los Angeles Lakers when they won the **winter cup finals** in **1826**...”

Design implication

We need semantic matching for paraphrases and lexical matching for exact facts.

Step 1: Dense Retrieval Baseline

Method

- all-MiniLM-L6-v2 whole-page embeddings, L2-normalized.
- Searched with FAISS IndexFlatIP (inner product = cosine similarity).
- Exact full vector search using one embedding vector per page.

Dense only - NDCG@10

0.329

- **Whole-page units beat windowed chunks.** Answer pages are short; windowing diluted the signal and sometimes let a long distractor win on one matching window.
Early test: 0.174 windowed vs. 0.224 whole-page.
- **Conclusion:** dense retrieval gives useful semantic recall, but it blurs exact facts such as years, populations, and coined names.
- **Example failure:** q2 asks for “about 1,456,779 residents”; embeddings alone do not reliably pin the exact number to the top.

Step 2: Lexical BM25 + Corpus Features

Okapi BM25

$$\text{RSV}^{\text{BM25}} = \sum_{i \in q} \log \frac{N}{df_i} \cdot \frac{(k_1 + 1)tf_i}{k_1 \left((1 - b) + b \frac{dl}{\text{avdl}} \right) + tf_i}$$

- k_1 controls term-frequency scaling.
- b controls document-length normalization.

BM25 only - NDCG@10

0.422

Recall@100

0.93

Tuned lexical retrieval

- Custom NumPy BM25, $k_1 = 2.0$, $b = 0.75$, with a term-major inverted index.
- Porter stemming: “negotiator” / “negotiations” → negoti. Raised recall@100 0.92 → 0.93.
- Decade tokens: 1826 and “1820s” both emit 182x.
- Word bigrams keep phrases such as “point guard” and “cold-water fisheries” together.
- Numeric tokens stay intact for exact population/year facts.

Why not lemmatization?

Lemmatization adds dependencies and POS tagging, which is heavy for runtime. Porter stemming is dependency-free and captures the main morphological variants.

Step 3: Hybrid Retrieval - Dense & Lexical

Fusion rule

$$s = 0.7 \cdot s_{\text{dense}} + 0.3 \cdot s_{\text{BM25}}$$

Hybrid ($\alpha = 0.7$) - NDCG@10

0.447

- Dense and BM25 scores are each **per-query min-max normalized** before fusion.
- Per-query normalization matters because cosine scores and BM25 scores live on different scales.
- We tuned end-to-end: $\alpha = 0.7$ slightly lowered pure hybrid compared with $\alpha = 0.6$ (0.447 vs. 0.450), but gave a more robust dense-leaning candidate pool for reranking.
- Dense recovers paraphrases; BM25 nails exact years, numbers, and coined phrases. Their complementary errors lift NDCG@10 above either signal alone.

Diagnosis: Recall Is High, Ordering Is the Bottleneck

Depth k	Mean Recall@ k	Oracle NDCG@10
10	0.59	0.60
20	0.75	0.80
50	0.89	0.91
100	0.93	0.94

- The gap is ordering, not recall.
- Gold pages are usually in the top-100 pool, and a perfect reranker could reach 0.94, yet the hybrid scores only about 0.45.
- Decision: rerank around 100-120 candidates. Deeper adds noise; shallower loses recall.

Step 4: Cross-Encoder Reranking

Reranker

- cross-encoder/ms-marco-MiniLM-L-6-v2 reads each (query, page) pair jointly and rescores the top fused candidates.
- Small model, about 80 MB, so loading and inference fit the GPU budget.
- Performed better than a larger BGE cross-encoder.

Reranker (blend 0.5) - NDCG@10

0.469

Blend, do not replace: CE helped single-answer queries, but hurt some multi-answer templated queries, so we blend:

$$\text{score} = 0.5 \cdot \text{reranker} + 0.5 \cdot \text{hybrid}.$$

Example: q0, “point guard ... 1820s”, was lifted to the top; one clean case jumped 0.63 → 1.00.

Results: Public 29 Queries

Configuration	NDCG@10	Δ
Dense only (whole page)	0.329	–
BM25 only (stemmed + decade + bigram)	0.422	+0.093
Hybrid: $0.7 \cdot \text{Dense} + 0.3 \cdot \text{BM25}$	0.447	+0.025
Cross-Encoder Reranker (blend 0.5)	0.469	+0.022
Final Optuna-tuned (blend 0.6, pool 120)	0.485	+0.016

Final public - NDCG@10

0.4854

Query phase time

28.06s

Robustness Check: Rephrased Queries

What we did & Why

- The public query set was small, so real cross-validation would be unstable.
- To further validate our results, we used Gemini 3.5 Flash to rephrase the given queries while preserving the intended meaning.
 - According to the forum, the given query set was also generated.
- We reused the original relevance labels, assuming the rephrased query still asked for the same pages.

New queries - NDCG@10

0.352

Interpretation: the score dropped, as expected, because BM25 loses some exact word overlap under new wording. The result is useful as a robustness check, but not an official validation set: an LLM rephrase may slightly change the intended meaning.

Example: “What river delta municipality has about 1,456,779 residents?” → “Which city located in a river delta has a population of approximately 1,456,779 people?”

- **Match the method to the corpus.** Short synthetic answer pages and long distractors drove our choices: whole-page units, BM25, and blending the reranker with the hybrid score.
- **Hybrid + rerank, each fixing a distinct failure.** Dense recovers paraphrase, BM25 nails exact facts, and the cross-encoder fixes ordering.
- **Ordering, not recall, was the bottleneck.** Recall@100 ≈ 0.93 and oracle NDCG@10 ≈ 0.94 versus hybrid ≈ 0.45 .
- **Small beat big; blend beat replace.** MS-MARCO L-6 transferred better than larger rerankers, and a hybrid prior rescued templated multi-entity queries.
- **Tuned carefully on a tiny set.** With only 29 queries, we adopted the robust region center rather than a noisy argmax.